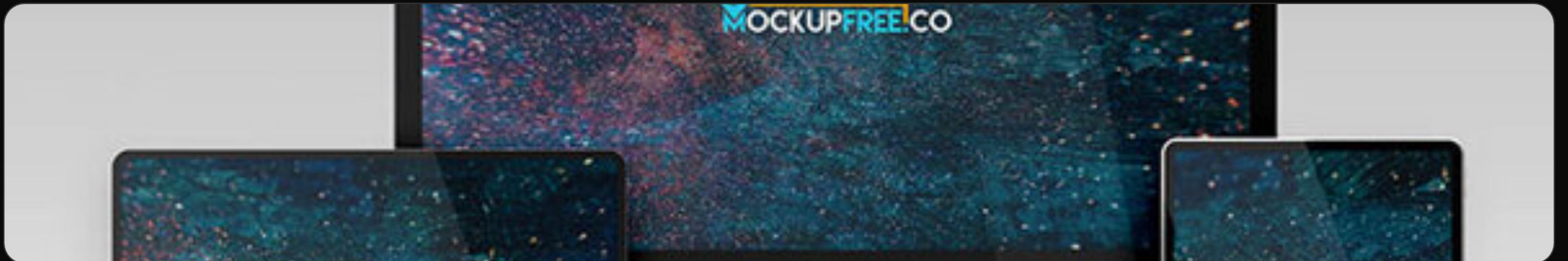


HANDHELD & WEARABLE COMPUTING

Exploring Module 2 & 3: Device Definitions, Touch Interfaces, and the Future of Interaction Design



CORE FOUNDATIONS

Defining the Technological Landscape of HCI

DEFINING THE MODERN DEVICE

TECHNOLOGICAL DEVICE

Any computing system, smartphone, or specialized electronic architecture engineered to create, store, or transmit information digitally.

USER INTERFACES (UI)

The space where interactions between humans and machines occur. The goal is effective operation and control of the machine from the human end.

GUI VS. TUI: THE SHIFT IN MODALITY



GUI

Graphical User Interface relies primarily on the sense of **sight**. It uses windows, icons, and menus driven by indirect input like a mouse.



TUI

Touch User Interface relies primarily on the sense of **touch**. It enables direct manipulation of digital objects via physical contact.



THE DIFFERENCE

While GUI users look and click, TUI users reach and touch, creating a more visceral and immediate sense of interaction.

TOUCHSCREEN TECHNOLOGIES PART 1

 Resistive vs Capacitive diagram

WIRE RESISTIVE

Activated by physical pressure from any object (finger, stylus, pen). Low cost but lower image clarity.

600 × 400

SURFACE CAPACITIVE

Requires human skin contact. Durable and better clarity, but limited to bare finger interaction.



PROJECTED CAP (P-CAP)

Supports multi-touch gestures. Highly sensitive and scratch-resistant, though susceptible to EMI/RFI.

HIGH-CLARITY OPTICAL SENSING

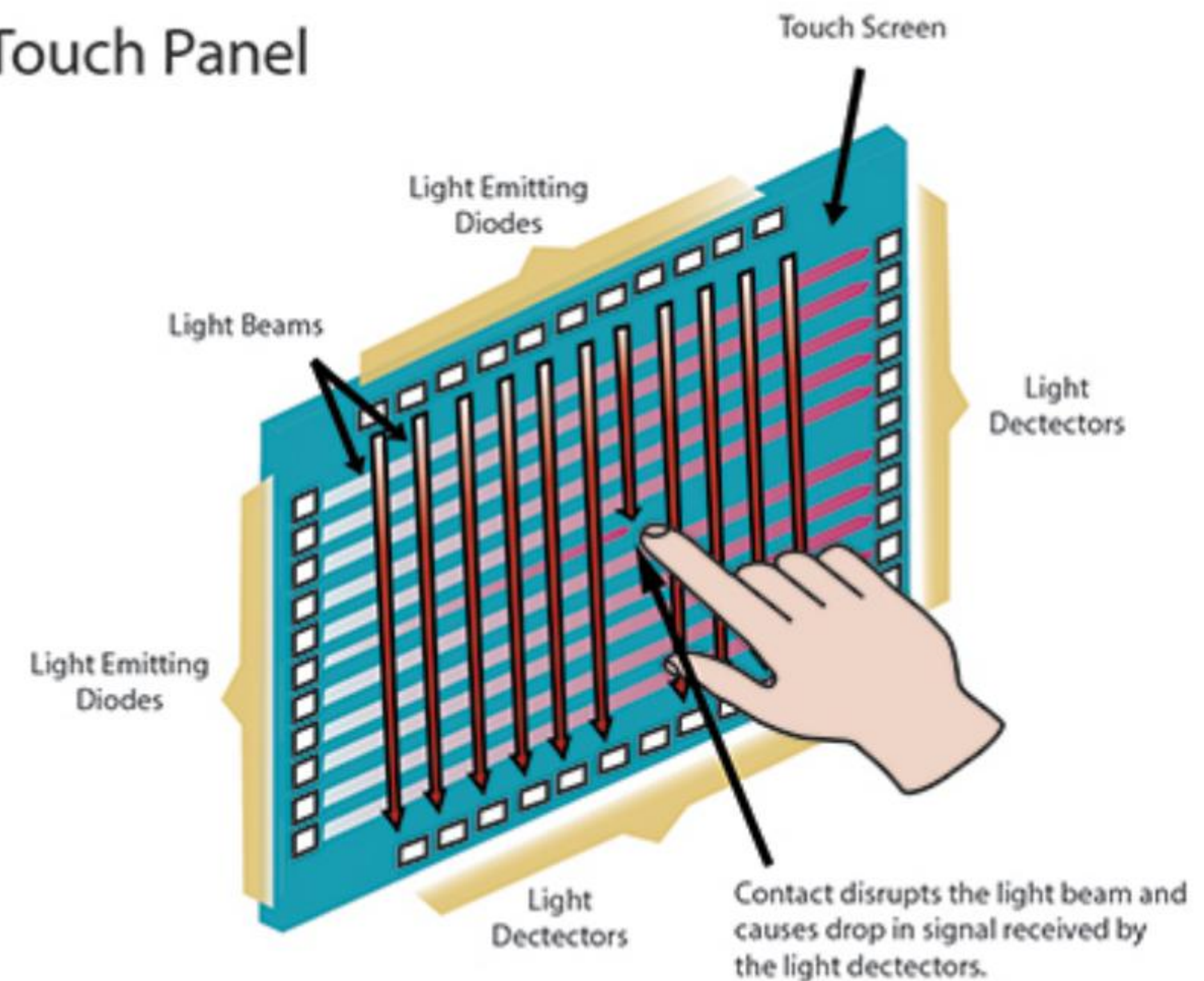
SAW (SURFACE ACOUSTIC WAVE)

Uses ultrasonic waves across the glass panel. Offers excellent clarity but can be disabled by solid contaminants or moisture on the surface.

INFRARED (IR) TOUCH

Uses an invisible grid of light beams. Offers the **highest possible clarity** and light transmission because no extra layers cover the display.

Infrared Touch Panel



EVOLUTION OF HANDHELD DEVICES

2000

Convergence

Threshold: Introduction of basic color screens and internet.

2009+

Smartphone Matrix: Universal displacement of PDAs by Android and iOS.

1985-1995

Genesis of Portability: First voice transceivers and bulkier mobile units.

2005

GPRS/3G Expansion: High-speed networks and high-resolution displays.



THE PIVOTAL SHIFT

2007

YEAR OF REINVENTION

MAINSTREAM MILESTONES

The simultaneous launch of the **Apple iPhone** and **Microsoft Surface** marked the moment touch-based interfaces entered the mainstream global market.



Original iPhone and Surface

CORE INTERACTION TECHNIQUES



TAPPING-IT

The most common gesture; selecting an item by briefly touching it. Fundamental to almost every modern UI action.



CROSSING-IN

Selecting a target by dragging across or crossing it. Useful in high-density interfaces where tapping might be inaccurate.



DIRECTIONAL GESTURING

Eliminates the strict requirement for predefined target spaces, allowing users to trigger actions based on motion paths.

WEARABLES & SMART TEXTILES

THE NEXT FRONTIER

Wearable technology is moving beyond watches to "Tech Textiles." These involve garments woven with built-in **biometric circuitry** to monitor health and performance in real-time.

From fitness trackers to AI-powered hearing aids and smart jewelry, the interface is becoming invisible and ubiquitous.

HCI DESIGN CONSIDERATIONS

- 🛡️ **Extension of the Body:** (Wigdor & Wixon, 2011) The device interface should feel like a natural extension of the user's physical self.
- 🌀 **Natural Experience:** Interaction should mimic real-world physics and intuitive human behaviors to reduce cognitive load.
- ⚡ **Immediate Feedback:** The system must respond instantly to touch to maintain the illusion of direct physical manipulation.
- 📏 **Size Factor:** Interfaces must account for ergonomic constraints and the physical size of the human hand and fingers.

QUESTIONS?

Assessment Review: HCI 2 - Module 2 & 3

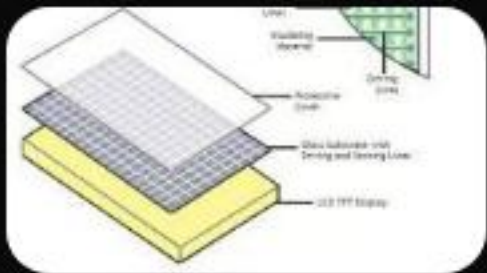
quiz content covered: touchscreen types, interaction history, and future wearables.

IMAGE SOURCES



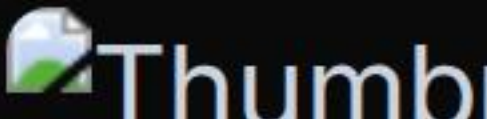
<https://resourceboy.com/wp-content/uploads/2022/01/5-device-set-mockups-including-laptop-smartphone-tablet-and-monitor-thumbnail.jpg>

Source: resourceboy.com



<https://global.discourse-cdn.com/digikey/original/2X/6/6e0443ff7f710841d21ea7e3659696b2f6793c00.JPG>

Source: forum.digikey.com



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Source: www.epectec.com



https://external-preview.redd.it/the-initial-cost-of-cell-phones-since-1984-adjusted-for-v0-vtTcuyGPBuUI4fLRIXICINts_49I-QeNBHNAOfzSBJY.jpeg?width=640&crop=smart&auto=webp&s=25518a36a5becd63a6ace991e1ddd21ba0ffcdbde

Source: www.reddit.com



<https://b2b-contenthub.com/wp-content/uploads/2024/10/Evolution-of-iphone-2024-01.jpg?quality=50&strip=all&w=1024>

Source: www.computerworld.com

IMAGE SOURCES



<https://www.shieldex.de/wp-content/uploads/2022/03/Smart-Clothes-und-Wearables-2000x2000-1.png>

Source: www.shieldex.de